OFFICIAL USER MANUAL

oXygen
MASTERING SUITE

oXygen Mastering Suite

Detailed operating guide for version 1.0.1, covering the full mastering chain, every module, automatic workflows, and a practical manual mastering process.

FORMAT

VST3

DEFAULT CHAIN

Graphic EQ -> Dynamic EQ -> Multiband Comp -> Stereo Imager -> Gain -> Maximizer

CORE USE

Final stereo bus mastering, mix bus finishing, or stem polishing when a controlled chain is needed.

This manual explains what every visible control does, what each module is for, how the automatic modes work, and how to use oXygen in a deliberate mastering workflow instead of blindly chasing loudness.

Table of Contents

1. What oXygen is and what it is for
2. Interface overview
3. Signal chain and processing order
4. Module-by-module reference
5. Master Assistant
6. Reference Match
7. Recommended manual mastering workflow
8. Starting points and style recipes
9. Quick troubleshooting
10. Final working philosophy

1. What oXygen Is

oXygen is a practical mastering plugin designed around a straightforward, ordered rack. It is intended to help you improve tonal balance, control build-up and harshness, add dynamic glue, shape stereo width, and finish with a controlled final loudness stage.

It is built primarily for stereo bus work. In most sessions that means the mix bus or the final master bus. It can also be used on stems, but its analysis and automatic tools are aimed first at full stereo program material rather than isolated instruments.

> **Recommended mindset:** correct first, control second, enhance third, and push level last. In mastering, a chain works best when each stage solves one specific problem instead of trying to do everything at once.

Main strength

Clear rack logic and automatic workflows that still write real parameters into the modules instead of hiding everything behind a black box.

Best placement

Insert on the final stereo bus after your mix is already fundamentally balanced. oXygen is for finishing and refining, not rescuing a broken mix.

Best results

Use the analyzer, trust your ears, and keep moves deliberate. Small, precise changes usually translate better than dramatic corrections.

2. Interface Overview

2.1 Top Bar

- **Logo area:** branding only.

- **REFERENCE:** loads an external `WAV` or `FLAC` file, analyzes it, then listens to your mix and builds a reference-driven mastering proposal.
- **MASTER ASSIST:** listens to up to 60 seconds of your own mix in real time and proposes a mastering setup based on the incoming audio.
- **Original vs Processed analyzer:** overlays the incoming curve and the processed curve so you can see whether your changes are balanced or excessive.

2.2 Side Meters

- **Input meter:** shows the level arriving at the plugin.
- **Output meter:** shows the level after the full rack.
- **Left / Right readout:** lets you spot obvious level imbalances between channels.

2.3 Rack Header Controls

- **B:** bypass the module.
- **Up / Down arrows:** move the module within the rack.
- **^:** collapse or expand the module view.

> **Important:** oXygen allows module order changes, but the stock order is already chosen for mastering logic. Unless you have a clear reason, keep the default chain intact.

3. Signal Chain and Why It Is Ordered This Way

1. **Graphic EQ**
   Handles broad tonal shaping before dynamic stages react.
2. **Dynamic EQ**
   Catches moving build-ups and harshness before compression glues them in.
3. **Multiband Comp**
   Adds control and glue once the tonal foundation is cleaner.
4. **Stereo Imager**
   Shapes width after tone and dynamics are more settled.
5. **Gain**
   Acts as internal trim before the final loudness stage.
6. **Maximizer**
   Sets final loudness and output ceiling at the end of the chain.

Stage	Main job	Why it sits there
Graphic EQ	Broad, static tonal shaping	It makes the rest of the chain respond to a more balanced signal.
Dynamic EQ	Conditional cuts on problem zones	It stops moving resonances from over-triggering the compressor.
Multiband Comp	Glue and band-specific control	It works more cleanly after tone and resonances are already tamed.
Stereo Imager	Width shaping by band	You can open the image more safely after dynamics are better behaved.
Gain	Internal trim	It feeds the limiter more intelligently instead of making the limiter solve everything alone.
Maximizer	Final loudness and ceiling protection	It always belongs at the very end.

MODULE ATLAS

How to read the rack

Every stage below is described in the order it appears in the chain, with its role, ranges, and the safest way to use it in a mastering context.

4. Module Reference

EQ

4.1 Graphic EQ

STATIC TONAL SHAPING

Graphic EQ is a 15-band fixed-frequency EQ designed for broad, musical mastering moves. It is the right place to fix overall darkness, excessive mud, a missing sense of presence, or too much top-end weight.

Internally it uses smoothed peak filters so parameter changes are not abrupt. The intent is not surgical repair but steady tonal steering.

Available bands

`30, 40, 60, 100, 180, 300, 500, 900, 1.5k, 2.5k, 4k, 6k, 10k, 15k, 20k`

What to listen for

- **30-60 Hz:** sub weight, rumble, low-end mass.
- **100-300 Hz:** body, punch, bloom, mud.
- **500-900 Hz:** boxiness, closed tone, nasal density.
- **1.5-4 kHz:** definition, bite, attack, presence.
- **6-20 kHz:** brightness, air, sheen, fatigue.

Control	Range	Meaning
Band Gain	-12 dB to +12 dB	Boosts or cuts energy around each fixed center frequency.

Recommended use: in mastering, +/-0.5 dB to +/-2 dB is already meaningful. If you need much larger moves, the issue is often in the mix rather than the master.

4.2 Dynamic EQ

CONDITIONAL CLEANUP

Dynamic EQ is for problems that appear only when the music gets dense, bright, or loud. Instead of applying a permanent cut, it waits until a band crosses its threshold and only then reduces it.

That makes it ideal for controlling mud, aggressive presence, splashy highs, or low-end surges without thinning the whole track all the time.

Band	Center	Typical target
LOW CLEAN	140 Hz	Low-end swell, kick or bass bloom.
BODY	380 Hz	Cloudiness, low-mid thickness, congestion.
PRESENCE	3.2 kHz	Edginess, bite, fatigue, hard snare or guitar presence.
AIR	9.6 kHz	Harsh cymbals, brittle top-end, splashy sheen.

Control	Range	Meaning	How to use it
TH	-48 dB to -6 dB	Trigger threshold. Lower values make the band act sooner and more often.	Lower it until the problem is controlled only when it appears.
RG	0 dB to 12 dB	Maximum dynamic cut allowed for that band.	Start around 1-3 dB. In mastering, 4-6 dB is already a strong move.

Best use case: if static EQ solves one section but damages another, Dynamic EQ is usually the better answer.

Warning: if Dynamic EQ feels inaudible, you may simply not be hitting the threshold. If it feels too obvious, the threshold is too low or the range is too large.

MB

4.3 Multiband Comp

GLUE AND CONTROL

Multiband Comp splits the signal into four bands and applies stereo-linked compression. The visible manual controls focus on what you most often need while mastering: crossover points, threshold per band, and output gain per band.

Ratio, attack, and release also exist internally, but in the current version those are mainly driven by the automatic engines instead of being front-panel manual controls.

Band roles

- **LOW:** sub and low-end stability.
- **L-MID:** weight, density, lower body, potential mud.
- **H-MID:** attack, forwardness, bite, upper energy.
- **HIGH:** top-end finish, cymbals, openness, brightness.

Control	Range	Meaning
Low-Mid Xover	20 Hz to 1000 Hz	Separates the low band from the low-mid band.
Mid-High Xover	500 Hz to 5000 Hz	Separates low-mid from high-mid.
High Xover	2000 Hz to 15000 Hz	Separates high-mid from high.
TH per band	-60 dB to 0 dB	Compression threshold for that band.
GN per band	-24 dB to +24 dB	Post-compression makeup or corrective gain.

Recommended use: multiband mastering compression should make the mix feel more stable and more glued, not smaller and flatter.

4.4 Stereo Imager

WIDTH BY BAND

Stereo Imager adjusts width separately in four frequency regions. In practice it scales the side component against the mid component, so it changes how wide a region feels rather than simply boosting left and right.

The safest mastering approach is to keep the lows controlled and do most widening in the upper mids and highs.

Control	Range	Meaning
Width Low / LowMid / HighMid / High	0.0 to 2.0	0.0 = mono, 1.0 = original width, 2.0 = extra wide.

- Keep **Low** close to 1.0 or below if mono solidity matters.
- Open **HighMid** and **High** first if the mix feels narrow.
- If the center starts to disappear, back off.

4.5 Gain

INTERNAL TRIM

Gain is part of the chain even though it does not currently have a dedicated manual editor panel in the rack. It serves as an internal level trim before the Maximizer.

This matters because a limiter works best when it is fed intentionally. If everything is pushed into the limiter with no prior level management, the result is usually smaller, harsher, and less controlled.

Control	Range	Meaning
Gain	0.0 to 2.0	Internal trim used by the automatic engines to feed the Maximizer more intelligently.

MX

4.6 Maximizer

FINAL LOUDNESS

Maximizer is the final loudness stage. It uses lookahead, adaptive release behavior, parameter smoothing, and 4x oversampling to make the limiting stage cleaner and more controlled than a raw sample-peak clip.

Its job is not only to make the master louder, but to decide how the track arrives at its final ceiling without losing impact more than necessary.

Control	Range	Meaning	Practical advice
Threshold	-60 dB to 0 dB	Lower values push more level into the limiter.	Bring it down slowly and listen for shrinking punch or brittle highs.
Ceiling	-60 dB to 0 dB	Maximum output ceiling.	For general music mastering, around -1.2 dB to -0.8 dB is a sensible starting point.
Release	1 ms to 500 ms	Limiter recovery speed.	Too short can sound hard. Too long can dull the mix and reduce movement.

Internal note: the module runs with 5 ms lookahead and 4x oversampling, which is why it behaves more like a proper finishing stage than a crude output clamp.

AUTO WORKFLOWS

Two different automatic entry points

Master Assistant listens only to your mix. Reference Match compares your mix to an external master and then writes a more reference-driven proposal into the same rack.

5. Master Assistant

Master Assistant is an automatic setup mode that listens to your own mix in real time and builds a starting configuration for the rack. It is not a preset loader. It writes actual values into Graphic EQ, Dynamic EQ, Multiband Comp, Stereo Imager, Gain, and Maximizer.

1. Listen

The plugin listens to up to 60 seconds of the incoming mix while your song is playing through it.

2. Analyze

It measures tonal balance, dynamics, transient behavior, stereo width, true peak, and loudness-related traits.

3. Propose

It converts the analysis into a complete rack suggestion.

4. Confirm

You approve the proposal before it is applied to the modules.

What it is good at

- Finding broad tonal imbalance.
- Detecting persistent low-mid build-up or harshness.
- Setting a safer starting point for loudness.
- Using Dynamic EQ as part of cleanup before glue and limiting.

What it still needs from you

- A representative section of the song.
- Manual judgment after the proposal is applied.
- A final decision about how aggressive the result should feel.

Best practice: feed Master Assistant the loudest or fullest meaningful section of the song, not an intro, breakdown, or quiet ambient passage.

6. Reference Match

Reference Match combines offline analysis of an external reference file with real-time listening of your own mix. Its goal is to move your master closer to the tonal profile, dynamic finish, stereo impression, and loudness behavior of the reference while still respecting safety guards against oversaturation.

1. Load

Select a `WAV` or `FLAC` reference file.

2. Analyze reference

The plugin extracts and analyzes a representative portion of the file internally.

3. Listen to your mix

It then listens to up to 60 seconds of your song playing through the plugin.

4. Match

It writes a proposal that prioritizes the reference profile, especially in EQ, while avoiding obvious clipping or oversaturation.

What Reference Match tries to copy

- Overall tonal shape and frequency weighting.
- Dynamic density and glue direction.
- Stereo width profile by region.
- Reasonable final loudness and ceiling behavior.

What it does not do

- It does not separate drums, guitars, vocals, bass, or stems individually.
- It does not replace a weak mix with a perfect master.
- It does not guarantee that the reference can be copied safely if the source mix is fundamentally different.

Current behavior: Reference Match gives strong priority to the reference profile, especially in EQ, but if the predicted result would become too stressed, it softens the proposal to avoid obvious clipping or destructive saturation.

MANUAL PLAYBOOK

Building a master by hand

This section is written as a repeatable workflow. Use it when you want to drive the chain yourself instead of relying on the automatic proposals.

7. Recommended Manual Mastering Workflow

The workflow below is for times when you want to master fully by hand. It is intentionally conservative and ordered. You can push harder later, but the best results usually come from first building a clean, controlled foundation.

Step 1. Preparation

- Insert oXygen on the stereo mix bus or master bus.
- Choose the loudest, busiest, or most revealing section of the song for decisions.
- Leave headroom if possible. Do not start by crushing the Maximizer.

Step 2. Shape broad tone with Graphic EQ

- Clean broad mud first, usually somewhere between 180 Hz and 500 Hz.
- If the mix lacks definition, explore small lifts around 2.5 kHz to 6 kHz.
- If the top is too shut-in, add air carefully at 10 kHz to 15 kHz.
- If the mix gets harsh, undo the boost and solve the problem with Dynamic EQ instead.

Step 3. Catch moving problems with Dynamic EQ

- Start with **RG** around 1-3 dB.
- Lower **TH** until the band acts only when the issue becomes obvious.
- Use LOW CLEAN for low-end surges, BODY for congestion, PRESENCE for bite, and AIR for brittle highs.

Step 4. Add glue with Multiband Comp

- Touch crossovers only if the band split clearly does not suit the mix.
- Use thresholds to tame the unstable band, not to flatten the whole track.
- Use band gain only to recover balance after control, not as a replacement for EQ.

Step 5. Refine width with Stereo Imager

- Keep the low band stable and centered.
- Open HighMid and High first if the song needs size or air.

- Check mono compatibility if the mix relies on club translation or strong center focus.

Step 6. Finish with Maximizer

- Set a sensible ceiling first, for example -1.2 dB to -0.8 dB.
- Lower threshold gradually until you reach the desired intensity.
- If impact collapses, cymbals turn brittle, or the mix shrinks, back off.
- Adjust release if the limiter feels too grabby or too slow to recover.

Step 7. Compare against the original

- Use the Original vs Processed overlay to check whether the changes make sense.
- Do not trust loudness alone. A louder master can seem better for the wrong reason.
- Take short breaks. Ear fatigue hides mistakes.

Manual mastering rule: if you only hear the change when the track is much louder, you may be chasing level instead of quality.

8. Starting Points and Style Recipes

Graphic EQ

Most useful mastering moves live in the +/-0.5 dB to +/-1.5 dB zone. Go further only when the mix clearly needs it.

Dynamic EQ

Begin with 1-3 dB range and lower threshold only until the issue is controlled during problem moments.

Multiband Comp

Use moderate thresholds and think in terms of stability, not audible pumping or dramatic reshaping.

Stereo Imager

Keep Low around 1.0 or below. Widen HighMid and High first, and only by as much as the center can still support.

Maximizer

Set ceiling first, then threshold. If you hear sharpness or lost punch, undo the last move before touching more modules.

Automatic modes

Treat Master Assistant and Reference Match as intelligent starting points. Apply, listen, then refine manually.

8.1 Style Recipes

These are not hidden plugin presets. They are practical starting directions you can dial in manually when you want to move quickly. Use them as tonal and dynamic intent, not as fixed rules.

Modern Pop / Indie Pop

Target feel: polished, open, controlled, center-forward vocals.

- Graphic EQ: trim a little mud around 180-300 Hz, add a slight presence lift around 2.5-6 kHz, then a gentle air lift above 10 kHz if needed.
- Dynamic EQ: keep BODY and AIR ready for congestion and splashy cymbals.
- Multiband Comp: moderate glue, especially in low-mid and high-mid.
- Stereo Imager: subtle widening in HighMid and High, conservative low band.
- Maximizer: aim for polish before loudness. Do not let brightness turn brittle.

Rock / Metal

Target feel: dense, punchy, controlled aggression, solid center.

- Graphic EQ: watch 180-500 Hz carefully; do not over-hype the top if cymbals are already busy.
- Dynamic EQ: BODY and PRESENCE are often the most useful bands.
- Multiband Comp: keep low-end and upper-mids stable without flattening drum attack.
- Stereo Imager: keep low and low-mid stable; widen upper regions only as much as the center can carry.
- Maximizer: preserve impact. If snare and kick lose authority, back off.

Trap / Hip-Hop

Target feel: deep low-end, clear vocal lane, controlled top-end, modern loudness.

- Graphic EQ: protect sub weight, clean low-mid masking, add presence only if the vocal lane feels buried.
- Dynamic EQ: LOW CLEAN and PRESENCE are often key.
- Multiband Comp: low band should feel stable, not strangled; high band should not spit.
- Stereo Imager: keep subs centered, open highs for atmosphere if the beat allows it.
- Maximizer: push confidently, but watch for smeared kick or crunchy hi-hats.

Electronic / Ambient

Target feel: wide, immersive, smooth, stable low-end foundation.

- Graphic EQ: broad tonal contour matters more than small corrective cuts.
- Dynamic EQ: use lightly unless a resonance or bright synth edge becomes distracting.
- Multiband Comp: aim for control and glue, not obvious pumping unless that is stylistically intentional.
- Stereo Imager: upper-band width can be more adventurous, but low-end still needs mono discipline.
- Maximizer: preserve space. If the mix feels smaller, the final stage is doing too much.

Acoustic / Singer-Songwriter

Target feel: natural, intimate, clear, unforced.

- Graphic EQ: small moves only; avoid over-brightening.
- Dynamic EQ: excellent for controlling occasional vocal bite or guitar resonance without flattening the performance.
- Multiband Comp: keep it subtle and supportive.
- Stereo Imager: usually less is more.
- Maximizer: stop before the recording loses breath and micro-dynamics.

General fallback recipe

Target feel: balanced, safe, translation-focused.

- Graphic EQ: one small cleanup move, one small enhancement move at most.
- Dynamic EQ: 1-2 bands active with low range.
- Multiband Comp: moderate thresholds, minimal gain correction.
- Stereo Imager: low near mono, highs slightly wider than source.
- Maximizer: ceiling first, threshold second, stop as soon as quality starts dropping.

9. Common Mistakes

Mistake	What usually happens	Better move
Boosting top end when the real problem is harshness	The mix sounds brighter at first but becomes fatiguing.	Use Dynamic EQ on PRESENCE or AIR instead of static boosting.
Using multiband compression to solve broad tonal imbalance	The mix gets flatter without becoming better balanced.	Correct tone first with Graphic EQ, then compress only what still needs control.
Widening low frequencies	Mono compatibility and center solidity get worse.	Keep the low band narrow and do most width work higher up.
Pushing the Maximizer before cleanup	Harshness, mud, and imbalances become more obvious and harder to control.	Fix tone and movement first, then push final loudness.
Trusting automatic modes without listening afterward	The proposal may be close, but not necessarily final.	Always compare, trim, and make the final calls yourself.

10. Quick Troubleshooting

If you hear this...	Check this first
The master still feels muddy	Graphic EQ around 180-500 Hz, then Dynamic EQ BODY.
The top end hurts when the song gets loud	Dynamic EQ PRESENCE or AIR before adding more loudness.
The mix loses punch after loudness processing	Ease off Multiband Comp and raise Maximizer threshold.
The center weakens after width changes	Reduce LowMid and HighMid width first.
The reference result feels too strong	Keep the reference match, then manually back off EQ, Dynamic EQ, or Maximizer pressure.

11. Final Working Philosophy

oXygen works best when every module has a clearly defined job:

- **Graphic EQ:** broad balance.
- **Dynamic EQ:** moving cleanup.
- **Multiband Comp:** glue and dynamic control.
- **Stereo Imager:** size and spread.
- **Gain and Maximizer:** final level management.

The best mastering results usually do not come from doing more. They come from doing the right amount, in the right place, in the right order. If the master feels clearer, more stable, more confident, and louder without sounding stressed, you are probably moving in the right direction.